

# Coordinate-Axis Points of the Complex Unit Ball Are Not MDM Points

Run `fa_banach.001`, lane 8

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## Status and source problem

**Status:** candidate counterexample, likely valid; human review requested.

In Definition 7 of their paper, PDF page 3, Kazaniecki and Wojciechowski introduce local and global MDM points and then conjecture that the polynomial-hull obstruction described there is the only obstruction to the MDM property [1]. They add that even for the Euclidean unit ball of  $\mathbb{C}^2$ , they do not know whether any MDM point exists. Here is the source passage.

Motivated by Theorem [5](#) we propose the following definition.

**Definition 7.** Let  $\Omega \subset \mathbb{C}^2$  and  $z \in \partial\Omega$  be its boundary point. We say that  $z$  is a local MDM point for  $\Omega$  provided there exists a ball  $B$  centered at  $z$  such that for every  $c > 0$ , there exists an analytic polynomial  $F$  on  $\Omega \cap B$  such that

$$\|F\|_{A(\Omega \cap B)} \leq 1, \quad \left\| \frac{\partial}{\partial z_1} F \right\|_{A(\Omega \cap B)} \leq 1, \quad \left\| \frac{\partial}{\partial z_2} F \right\|_{A(\Omega \cap B)} \leq 1$$

and

$$\left\| \frac{\partial^2}{\partial z_1^2} F \right\|_{A(\Omega \cap B)} \leq 1, \quad \left\| \frac{\partial^2}{\partial z_2^2} F \right\|_{A(\Omega \cap B)} \leq 1$$

but

$$\left| \frac{\partial^2}{\partial z_1 \partial z_2} F(z) \right| \geq c.$$

Similarly for bounded domains  $\Omega$  we define the global MDM point by replacing  $B$  by  $\mathbb{C}^2$  in the above definition. Using this notion we can reformulate Theorem [5](#) saying that point  $\mathbf{1} := (1, 1)$  is a (global) MDM point of the unit bi-disc (actually it is easy to see that any Shilov boundary point of a bi-disc is its MDM point). Note that MDM property is inherited by inclusion i.e. if  $z \in \overline{\Omega} \subset \overline{\Omega'}$  is an MDM point of  $\Omega'$  then  $z$  is an MDM point of  $\Omega$ . There are some natural restrictions on  $z \in \partial\Omega$  to be an MDM point. Indeed, suppose  $z \in \partial\Omega$  is an internal point of  $\hat{\Omega}$  - the polynomial convex hull of  $\overline{\Omega}$ . Then there exists a ball  $B$  centered at  $z$  contained in  $\hat{\Omega}$ . For every analytic polynomial  $f$  we have

$$\left| \frac{\partial^2}{\partial z_1 \partial z_2} f(z) \right| \leq C_B \sup_{\omega \in B} |f(\omega)| \leq C_B \sup_{\omega \in \Omega} |f(\omega)|.$$

The first inequality is a consequence of the Cauchy formula [\[14\] 1.2.2. \(v\)\]](#) and second follows by definition of polynomial convex hull [\[6\] p. 53\]](#). We conjecture that above obstacle is the only one which prevents the MDM property. However even in the case of the unit ball of  $\mathbb{C}^2$  we do not know whether any MDM point exists.

For clarity, put

$$\mathbb{B}^2 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 < 1\}.$$

In the source definition, a boundary point  $p$  is *local MDM* if some Euclidean ball  $B(p, R)$  has the following property: for every  $c > 0$ , an analytic polynomial  $F$  has all five norms

$$\|F\|, \quad \|F_1\|, \quad \|F_2\|, \quad \|F_{11}\|, \quad \|F_{22}\|$$

at most one on  $\mathbb{B}^2 \cap B(p, R)$ , while  $|F_{12}(p)| \geq c$ . The global definition replaces the cap by all of  $\mathbb{B}^2$ . Subscripts denote complex coordinate derivatives.

## Idea of the proof

At  $p = (1, 0)$ , restrict  $F_2$  to the axial disc and write  $h(\zeta) = F_2(\zeta, 0)$ . The transverse fiber of  $\mathbb{B}^2$  above a real point  $r$  is the disc of radius  $\sqrt{1 - r^2}$ . Cauchy's estimate on that fiber, applied to  $F_{11}$ , yields

$$|h''(r)| \leq \frac{1}{\sqrt{1 - r^2}}.$$

This singularity is integrable at  $r = 1$ , so  $h'(1) = F_{12}(1, 0)$  is bounded. Near  $p$ , every sufficiently late full transverse fiber lies in an arbitrary prescribed cap, which upgrades the global argument to a local one. Phase rotations and coordinate interchange then give all coordinate-axis boundary points.

## Counterexample theorem

**Theorem 1.** *For every  $\theta \in \mathbb{R}$ , neither*

$$p_\theta = (e^{i\theta}, 0) \quad \text{nor} \quad q_\theta = (0, e^{i\theta})$$

*is a local or global MDM point of  $\mathbb{B}^2$ .*

*More quantitatively, if  $F$  is a polynomial satisfying*

$$\|F_2\|_{L^\infty(\mathbb{B}^2)} \leq 1, \quad \|F_{11}\|_{L^\infty(\mathbb{B}^2)} \leq 1,$$

*then*

$$|F_{12}(1, 0)| \leq 1 + \frac{\pi}{2}.$$

*For every  $R > 0$ , there is also a finite constant  $C_R$  such that the same endpoint derivative is at most  $C_R$  whenever these two bounds hold on  $\mathbb{B}^2 \cap B((1, 0), R)$ .*

*Proof.* We first prove the global estimate at  $p = (1, 0)$ . Define

$$h(\zeta) = F_2(\zeta, 0), \quad \zeta \in \mathbb{D}.$$

Then  $|h| \leq 1$  on  $\mathbb{D}$ , and Cauchy's derivative estimate on discs of radius tending to one gives

$$|h'(0)| \leq 1. \tag{1}$$

For each real  $0 \leq r < 1$ , the function

$$g_r(w) = F_{11}(r, w)$$

is holomorphic and bounded by one on  $|w| < \sqrt{1-r^2}$ . Since  $g'_r(0) = F_{112}(r, 0) = h''(r)$ , another Cauchy estimate gives

$$|h''(r)| \leq \frac{1}{\sqrt{1-r^2}}. \quad (2)$$

Integrating along the real radius and using (1)–(2),

$$|h'(r)| \leq 1 + \int_0^r \frac{dt}{\sqrt{1-t^2}} \leq 1 + \frac{\pi}{2}.$$

Because  $F$  is a polynomial,  $h'$  is continuous at one. Letting  $r \uparrow 1$  proves

$$|F_{12}(1, 0)| = |h'(1)| \leq 1 + \frac{\pi}{2}. \quad (3)$$

The source normalization is stronger than the two hypotheses used in (3), so  $(1, 0)$  cannot be a global MDM point.

We next establish the cap-local assertion. Fix  $R > 0$ , and suppose only that  $|F_2| \leq 1$  and  $|F_{11}| \leq 1$  on

$$U_R = \mathbb{B}^2 \cap B((1, 0), R).$$

Set

$$\varepsilon = \min \left\{ \frac{1}{8}, \frac{R}{4}, \frac{R^2}{8} \right\}, \quad r_0 = 1 - \varepsilon, \quad \rho = \frac{\varepsilon}{2}.$$

If  $|\zeta - r_0| < \rho$ , then

$$|\zeta| < 1, \quad |1 - \zeta| < \frac{3\varepsilon}{2} < R.$$

Thus this entire axial disc lies in  $U_R$ . Cauchy's estimate for  $h$  at its center gives

$$|h'(r_0)| \leq \frac{1}{\rho} = \frac{2}{\varepsilon}. \quad (4)$$

For  $r_0 \leq r < 1$  and  $|w| < \sqrt{1-r^2}$ , one has

$$|(r, w) - (1, 0)|^2 = (1-r)^2 + |w|^2 < (1-r)^2 + (1-r^2) = 2(1-r) \leq 2\varepsilon < R^2.$$

Consequently the *full* transverse fiber above every such  $r$  lies in  $U_R$ , and the proof of (2) applies unchanged. From (4),

$$\begin{aligned} |F_{12}(1, 0)| &= |h'(1)| \\ &\leq \frac{2}{\varepsilon} + \int_{r_0}^1 \frac{dt}{\sqrt{1-t^2}} = \frac{2}{\varepsilon} + \arccos(r_0) =: C_R < \infty. \end{aligned} \quad (5)$$

Since (5) holds for every  $R > 0$ , no ball centered at  $(1, 0)$  can support the arbitrarily large mixed derivatives required by the local MDM definition.

Finally, if  $p_\theta = (e^{i\theta}, 0)$ , apply the result at  $(1, 0)$  to

$$G(z_1, z_2) = F(e^{i\theta} z_1, z_2).$$

The ball and corresponding caps are preserved, all relevant norms are unchanged, and  $|G_{12}(1, 0)| = |F_{12}(p_\theta)|$ . This proves the claim for  $p_\theta$ . Interchanging the two coordinates converts the hypotheses used above into the source bounds on  $F_1$  and  $F_{22}$ , and proves the claim for every  $q_\theta$ .  $\square$

## Why this disproves the proposed obstruction criterion

The closed Euclidean unit ball is its own polynomial hull. Indeed, if  $z \notin \overline{\mathbb{B}^2}$ , then the holomorphic linear polynomial

$$L_z(w) = \overline{z_1}w_1 + \overline{z_2}w_2$$

satisfies

$$|L_z(z)| = \|z\|_2^2 > \|z\|_2 = \sup_{w \in \overline{\mathbb{B}^2}} |L_z(w)|.$$

Thus  $p_\theta, q_\theta$  lie on, not in the interior of, the polynomial hull. Theorem 1 nevertheless says that they are not MDM points. This is a direct counterexample to the conjecture that hull interiority is the only obstruction.

## Verification notes

The proof is analytic and does not depend on computation. The reusable script `code/verify_axis_bound.py` checks the antiderivative in (3), the choice of  $\varepsilon$ , and all strict cap inequalities for 15 positive radii ranging from  $10^{-6}$  to  $10^3$ , together with several intermediate values. It reports

$$1 + \pi/2 = 2.570796326795 \dots$$

and that all transcription checks pass. The command used was

```
conda run --no-capture-output -n sandbox python code/verify_axis_bound.py
```

These checks are not a substitute for the Cauchy-estimate proof.

## Limitations and failed full-classification upgrade

The theorem completely classifies the coordinate-axis boundary orbit under phase rotations and coordinate interchange, but it does not decide whether a generic boundary point  $(a, b)$ , with  $ab \neq 0$ , is MDM. Along the analogous coordinate slice at such a point, the available transverse radius is only of order  $1 - r$ . Direct Cauchy estimation therefore produces a  $(1 - r)^{-1}$  singularity, which is not integrable. A general unitary rotation does not repair this because the source normalization controls the coordinate-pure derivatives  $F_{11}, F_{22}$ , a family not preserved by general unitary mixing. Accordingly, this packet does not claim that the unit ball has no MDM points.

## Bounded novelty check

The search performed through August 11, 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; the local source corpus; and bounded official-arXiv searches using `arXiv:2301.09526`, the exact title, both author names, “MDM point”, “Mitiagin-DeLeeuw-Mirkhil”, and “unit ball”. No duplicate packet or later explicit resolution of this unit-ball question was found. This is a bounded novelty check, not a claim of priority.

## Human review recommendation

A reviewer should check (i) the identification  $h''(r) = F_{112}(r, 0)$ , (ii) the two Cauchy estimates, (iii) the full-fiber cap inclusion preceding (5), and (iv) that the source's phrase "the above obstacle is the only one" is intended to cover the global MDM notion. The generic-boundary limitation above should remain explicit.

## References

- [1] K. Kazaniecki and M. Wojciechowski, *On the analytic version of the Mitiagin-DeLeeuw-Mirkhil non-inequality on bi-disc*, arXiv:2301.09526, 2023.